

Development Of Low Cost Stereolithography Based Resin Printer

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This research paper demonstrates the three different prototypes used in the development of a resin 3D printer. The goal of this research was to develop a dependable and effective 3D printer that could produce intricate objects with high-quality resin-based materials. Each prototype signifies an important development in the eventual product's functionality and design.

Prototype 1, which includes 3-axis simulation system, served as an important starting point. The team was able to assess the printer's operating principles and pinpoint areas for development through real-time simulations. This prototype made it easier to comprehend the technical details related to resin 3D printing.

To further improve the printer's accuracy and control, Prototype 3 added a one-axis stepper motor setup. In this setup, resin was contained in a vat tank, and a movable platform was moved precisely by an Arduino-based system. A digital micrometer of special design was also used to adjust the layer height. By making these improvements, the printing process was intended to be more accurate.

Keywords: Resin 3D printer, Prototype development, Additive manufacturing (AM), Liquid photopolymer, IOT Simulations, Arduino-based system, Accuracy, and control.

1 Introduction

A procedure for creating 3D objects known as additive manufacturing (AM), also known as 3D printing, involves successfully assembling material layers under the guidance of a computer program to create a physical object. The 3D file source is typically broken up into several layers, with each level providing its own unique set of computer-controlled instructions. The sequential-layer material addition or joining that takes place during a 3D production demonstrates the similarities between additive manufacturing and 3D printing. Direct and indirect 3D printing technologies can be separated into two groups. The main difference is whether the model was created via 3D printing or if the design was created directly using this technology. [1]

Photopolymerization a heat source selectively cures a liquid photopolymer contained in a vat (or tank), allowing a 3D physical object to be constructed layer by layer until it is complete. A variety of alternative techniques using 3D printing technology are based on the same basic idea. Aside from the first technique, which is based on lasers, there are numerous types of curing apparatus. LCD screens, Digital Light Processing projectors, and even Stereo Litho Graphy (SLA) are currently popular ways for photopolymerizing materials due to their low cost and exceptionally high resolution. The advantage of these three techniques over lasers is that they may simultaneously cure a complete resin layer, as opposed to the laser, which must gradually illuminate the entire surface by dragging it. [1][2][3]

2 Methodology

Technically known as thermoset polymers, resins utilized in DLP and SLA printing can be either rigid or flexible. Others are advertised as castable resins that can burn out when used as patterns during investment casting, while some are promoted as high strength engineering resins.

In comparison to extrusion operations, resin 3D printing techniques may generate more high-quality items with superior resolution and surface smoothness.

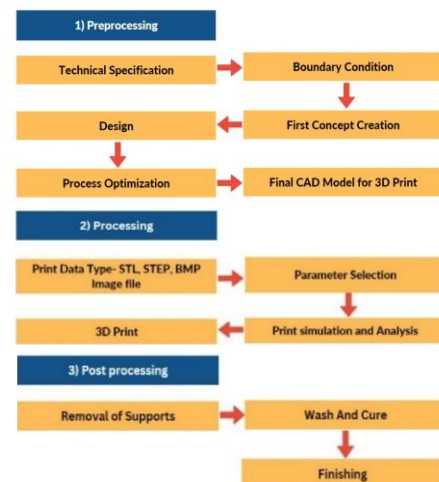


Figure 1. Flowchart of general SLA Printers [7] [8] [23]

ADVANTAGES: Excellent detail and quality. Compared to molten plastic, they are simpler to shape into intricate forms. The polymerization step creates prints with a stronger

structure and a smoother surface. a variety of materials are available. Even though there are just a few different kinds of resins, they may nevertheless be used for all essential purposes and meet all necessary specifications. Finished items with unique functionality can be created using various resin mixes. Quick motion. Resin printing is quicker than the prototype approach. Resin printers can create intricate pieces without compromising quality in a few hours or a day, depending on the size.

LIMITATIONS: Because resin parts are frequently fragile and frail, there aren't many uses for them. Large-scale objects can't always be printed all at once because it would be exceedingly expensive. Ordinary printers are not large because resin printing is used to create smaller and more precise models. [5]

2.1 Component Selection

- i. 1 x Arduino UNO
- ii. 1 x ST7735 LCD 1.8"
- iii. 1 x SD card module
- iv. 30 x LEDs (UV violet)
- v. 1 x TIP31 BJT transistor
- vi. 1 x NEMA17 Step motor
- vii. 1 x TMC2209 Driver
- viii. 1 x End-stop switch
- ix. 1 x micro-SD card
- x. 1 x 5W 15-ohm resistor

2.2 Proposed system Setup

2.2.1 Prototype 2



Figure 2. CAD Model – Prototype 2 – Wooden box basic Setup

2.2.2 Prototype 3

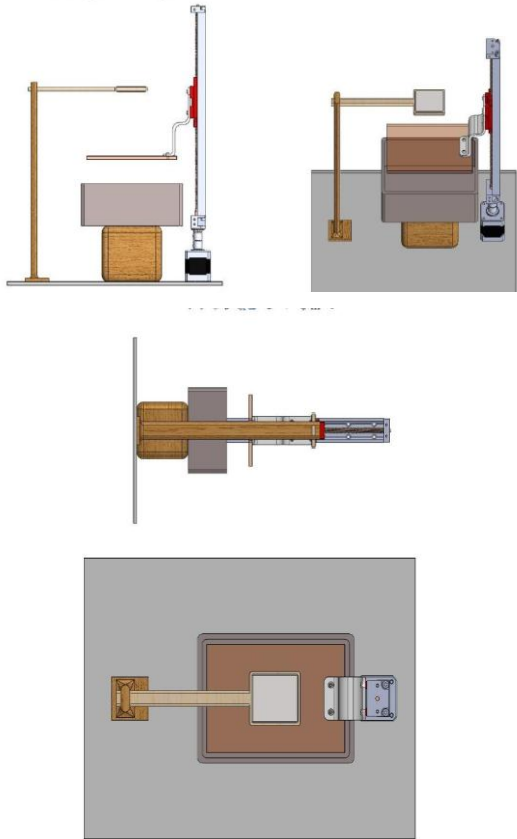


Figure 3. CAD Model – Prototype 3

2.3 Calculations

Tank calculations for minimum height
 Designing and developing a resin 3D printer involves several calculations to ensure accurate printing, optimal performance, and safety considerations. [5][7][8]

1. Print Volume and Build Plate Size:
 Given the dimensions of the components as 35 cm x 25 cm x 13 cm, the volume of the components can be calculated as:

$$\text{Volume} = \text{Length} \times \text{Width} \times \text{Height}$$

$$= 35 \text{ cm} \times 25 \text{ cm} \times 13 \text{ cm}$$

$$= 11,375 \text{ cm}^3 \dots\dots\dots (1)$$

• **Total Resin Volume:**
 Considering the dimensions of the resin tank as 42 cm x 43 cm x 16 cm, the total volume of resin required can be calculated as:

$$\text{Total Resin Volume}$$

$$= \text{Length} \times \text{Width} \times \text{Height}$$

$$= 42 \text{ cm} \times 43 \text{ cm} \times 16 \text{ cm}$$

$$= 28,896 \text{ cm}^3 \dots\dots\dots (2)$$

2. Total Volume with Resin:
 Calculating the total volume required for the resin 3D printer, including the components and resin, we add the volume of the components to the total resin volume:

$$\text{Total Volume with Resin}$$

$$= \text{Volume of Components} + \text{Total Resin Volume}$$

$$= 11,375 \text{ cm}^3 + 28,896 \text{ cm}^3$$

$$= 40,271 \text{ cm}^3 \dots\dots\dots (3)$$

3. Required Height of Tank:

Determining the required height of the resin tank, we divide the total volume with resin by the area of the tank's base:

Required Height of Tank

$$= \text{Total Volume with Resin} / (\text{Length} \times \text{Width})$$

$$= 40,271 \text{ cm}^3 / (42 \text{ cm} \times 43 \text{ cm})$$

$$\approx 22.218 \text{ cm}$$

$$\approx 25 \text{ cm (rounded to the nearest whole number)} \dots\dots\dots (4)$$

4. Platform Movement and Resolution:

Determining the required steps per millimeter or degree for the stepper motor driving the platform movement. This calculation depends on the desired resolution of the printer, which affects the layer height and the precision of the printed object.

The parameters we have selected in our goal to achieve is 300 mm/s. Let's consider our printer have an acceleration setting at 3000 mm/s².

$$\text{Time to Accelerate} = \text{speed} / \text{acceleration}$$

$$= (300\text{mm/s}) / (30000\text{mm/s}^2)$$

$$= 0.1\text{s.} \dots\dots\dots (5)$$

Knowing that our printer takes 0.1 second to reach its full speed isn't very useful. However, we can find the minimum distance for maximum speed easily.

Minimum distance

$$= \frac{1}{2} * 3000 * 0.1^2$$

$$= 15\text{mm.} \dots\dots\dots (6)$$

15mm is quite long for a single edge. This distance is for accelerating only. The worst case, like a square edge, will require decelerating too, so the same length will be required. In short, we need to print a 30mm square if we want to reach 300 mm/s during a fraction of a second. In this case, the average speed will be about half of the max speed, so 150mm/s.

5. Ball Screw and Motor Selection: Calculating the appropriate ball screw pitch and diameter based on the required platform movement and the torque generated by the stepper motor. We will also ensure that the ball screw can handle the load and provide smooth and precise movement.

- Ball screws are less common in 3D printers but more common in CNC. It has a higher mass, thus a higher inertia, and might be bottleneck in a 3D printer. Let's see what would happen on a Y axis.
- Most economic ball screws are 12mm in diameter and have a pitch of 4mm.
- Since these kinds of screw are very efficient, the torque required to move a mass of 1015 grams can be quite low.

$$\text{Load Torque} = (\text{pitch} * \text{friction} * \text{mass} * \text{gravity}) / 2\pi$$

$$= (4\text{mm} * 0.10 * 1.015\text{kg} * 9807\text{m/s}^2) / 2\pi$$

$$= 0.633\text{N.m} \dots\dots\dots (7)$$

The rolling balls generate very little friction and the lead pitch act as a reduction ratio compared to a belt system. For a full turn, the ball screw will rotate 4mm.

6. Motor Torque: Considering factors such as the weight of the platform, the height of the printing area, and any

additional forces or resistance in the system we have selected- Stepper motor NEMA 17 with

$$\text{Holding Torque (Kg-Cm)- } 4.2 \text{ (Kg-Cm)} \dots\dots\dots (8)$$

$$\text{7. Power Supply: } 12\text{V} \dots\dots\dots (9)$$

2.4 Resin Selection and Material research analysis

Stereolithography (SLA) is a type of 3D printing process that uses a laser to solidify layers of liquid resin to create three-dimensional objects. The selection of resin for SLA plays a critical role in achieving the desired mechanical properties, surface finish, and overall quality of the printed parts. Here is a detailed description of resin selection for stereolithography:

- Material properties:** The first factor which is considered while selecting a resin is the mechanical factor. This includes mechanical properties such as strength, stiffness, toughness, and flexibility. These properties are determined by the chemical structure of the resin and its molecular weight, higher the molecular weight better the mechanical properties of the resin.
- Optical properties:** Optical properties are an important aspect which are considered while selecting the resin. It includes the transparency, colour, and ability to transmit light of the resin. The resin should have the desired level of transparency or opacity depending on the application. The final appearance of the 3d printed parts depends on the colour of the resin.
- Viscosity:** Printing process is affected by the viscosity of the resin. Resins with higher viscosities can be more difficult to print with but can produce parts with better surface finish and detail. Resins with lower viscosities are easier to print with but may produce parts with lower quality surface finish and detail.
- Curing mechanism:** There are two types of curing mechanisms for SLA resins: photopolymerization and thermal curing. Photopolymerization involves curing the resin using a UV laser, while thermal curing involves using heat to cure the resin. The choice of curing mechanism will depend on the desired properties of the final part and the printing process.
- Post-curing requirements:** Some resins require post-curing to achieve the desired mechanical properties. This involves exposing the printed part to UV light or heat to complete the curing process. The post-curing time and temperature will vary depending on the resin used.
- Safety considerations:** Resin selection should also consider safety considerations. Some resins may be toxic, flammable, or emit harmful fumes during printing or curing. It is important to follow safety guidelines and use appropriate personal protective equipment when working with SLA resins.

2.4.1 Features and benefits:

- Value for money
- Ultra-fast
- Easy to print.
- High resolution
- Ultra safe, without any odour
- Excellent printability with most SLA, DLP & LCD printers

2.4.2 TDS and SDS analysis

A Technical Data Sheet (TDS), a Safety Data Sheet (SDS) and samples are crucial for buyers to determine if the quality of your product is suitable for them. You can use this workbook to improve your access to the European food ingredients market.

The GP Resin cures quickly resulting in high printing speeds (layer curing as fast as 500ms with some systems) and smooth surface finish.

- GP Resin has relatively low viscosity for effortless cleaning, material recovery and manipulation yet having zero Volatile Organic Compounds.
- The reactivity of the resin has been customized for precision even when the reaction rate is fast. Supports are easy to remove, and part can be finished with standard tools used for buffering. [9][10][11]

Sr. No	Mechanical Properties	Post Cured resin
1.	Density (g/cm3)	1.3
2.	Experimental shrinkage (%)	1.8
3.	Elongation @peak (%)	3.5
4.	Young’s modulus(mpa)	1088
5.	Tensile stress @ break (mpa)	28
6.	Shore d hardness	81D
7.	Viscosity	20

Table 1. TDS analysis parameters

Safety data sheet analysis

- BASIC RESIN is a liquid monomer for use with DLP and LCD 3D Printer.
- Mechanical properties like PLA and thermal properties like ABS, which combines easy printability, high thermal stability, and stiffness.
- Printing parameters Specimens printed on an Anycubic Photon 1 at 23°C and 50% humidity 0.05mm layer height.
- 10 seconds exposure time per layer
- Post curing parameters 30min with 200W 405nmUV LED Conditioned 72h at 23°C and 50% humidity [9][10][11]

2.5 Prototype Analysis

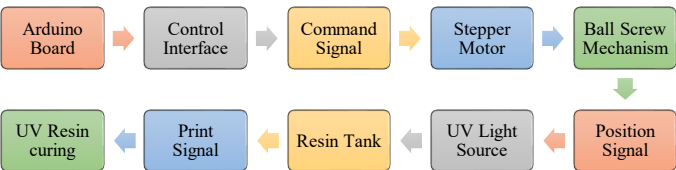


Figure 4 Flowchart of Circuit

Circuit Flow: The circuit flow within the resin 3D printer follows a sequence of operations to facilitate successful 3D printing:

- The user inputs commands through the control interface, providing instructions on the object design and print settings.
- The Arduino board receives these command signals and processes them accordingly.
- Based on the received commands, the Arduino sends position signals to the stepper motor, initiating the movement of the platform in the desired direction and distance.
- Simultaneously, the Arduino triggers the UV light source, sending print signals to control the curing process.
- As the platform moves, the UV light source emits UV radiation onto the resin, curing each layer and building the object from bottom to top.
- This process continues until the entire object is printed, following the design specifications provided by the user.

2.5.1 Prototype 1- Simulation

Proteus is a widely used software tool for simulating electronic circuits and systems, including microcontroller-based devices like Arduino. In the context of an Arduino-based resin 3D printer, Proteus simulation was crucial for validating circuitry and ensuring printer functionality before physical implementation. By emulating hardware components and control signals, it allowed verification of circuit connections, testing of Arduino board responsiveness, and assessment of overall printer performance virtually. The simulation involved creating a virtual model of the Arduino board and associated components, defining inputs/outputs, configuring control interfaces, and establishing circuit connections. Simulated commands were input through the control interface to observe Arduino board responses, including controlling stepper motors, activating UV light sources, and managing resin curing. Evaluation parameters included movement accuracy, synchronization between components, and overall printing process coordination. Real-time monitoring of various parameters and signals such as position and print signals was also possible during the simulation.

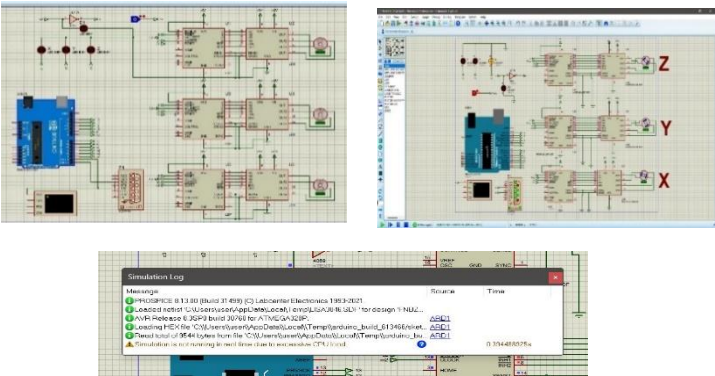


Figure 5 Proteus circuit Simulation and Run time Log.

2.5.2 Prototype 2 - Testing of UV light chamber setup to create test models.

In this prototype we have made a wooden chamber where we have connected a UV light at the inner top of the roof of the box. The chamber consists of a tray which can be placed for printing process. The chamber is divided into 2 parts where the smaller portion is used for component placing. We have given it a separate compartment to protect the components from UV rays. The tray is used to place the components on which we need to print the resin.

- Prototype manufacturing for box setup with UV lamp compartment and circuit compartment:



Figure 6. Wooden Fabrication

1. Material Selection: We chose suitable wooden sheets for constructing the box and ensured they are sturdy, durable, and of appropriate thickness.
2. Box Dimensions:
Height: 50 cm
Length: 40 cm
Width: 40 cm
3. Box Construction: To create the box structure, we cut the hardwood sheets to the specified proportions and put them together.
4. UV Lamp Compartment: The UV lamp compartment is located in the upper portion of the box. With the aid of suitable fixtures or brackets, the UV lights are firmly fastened to the top of the compartment. To allow heat from the UV lamps to dissipate, we put slits or ventilation holes on the back of the chamber.
5. Circuit Compartment: A partition was added between the compartment for the UV lamp and the smaller compartment for the circuit setup to avoid the UV light from influencing the circuit. The partition is constructed of non-conductive wood. To avoid any electrical interference or risks, we have secured sufficient insulation and grounding.
6. Quality Control: We have thoroughly inspected the finished box to look for any structural flaws, loose connections, or potential risks. We also made sure that all electrical connections and components were installed and secured in the correct manner.

2.5.3 Prototype 3 - Digital micrometer interfacing in the set up using stepper motor in Y axis.

i. Circuit explanation

We have used I2C Protocol in this circuit to establish seamless connection between the LCD Display and Arduino. The inter-integrated controller is referred to as I2C. This serial communication protocol can link devices that operate at low speeds. Multiple slaves can be connected to and controlled by a single master in this master-slave communication system. Each slave device in this has a unique address.

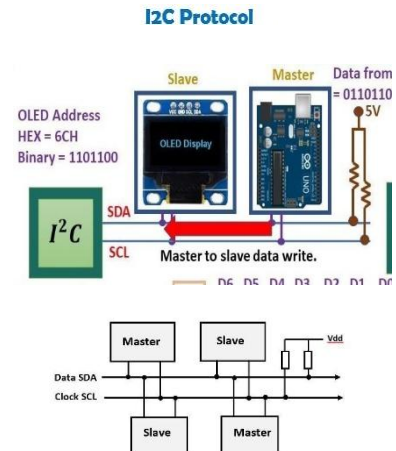


Figure 7. I2C Protocol [22]

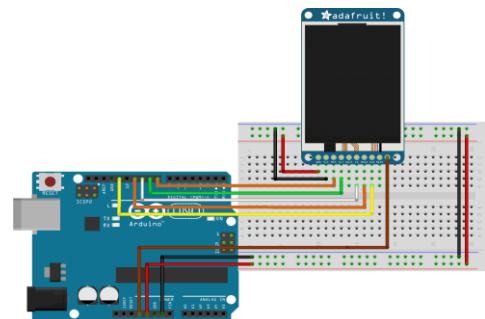


Figure 8. Arduino Circuit [22]

- Establish the connections, then go to the next section to obtain the code and libraries.

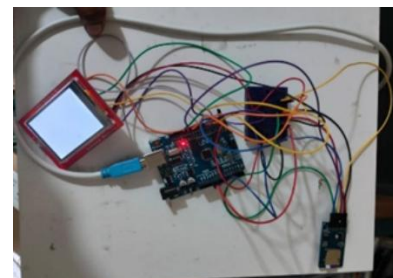


Figure 9. Connection circuit

In this prototype we have used a stepper motor of y-axis to move the platform vertically in the y-axis. We will be placing the LCD display in such a way that the displayed image will be printed on the resin platform. The resin tank will be placed right below the LCD display. The platform is constructed in such a way that it can be submerged into the tank and can collect a layer of resin on it.



Figure 10. LCD Output

Sr No	Parameter	Value
a)	Wavelength	365 nm
b)	Critical Energy (Ec)	8.7 J/mm ²
c)	Cure Depth (Cd)	1 cm
d)	Depth of Penetration (Dp)	0.961 mm
e)	Irradiance (Irr)	8 W/m ²

ii. Set up for prototype 3:

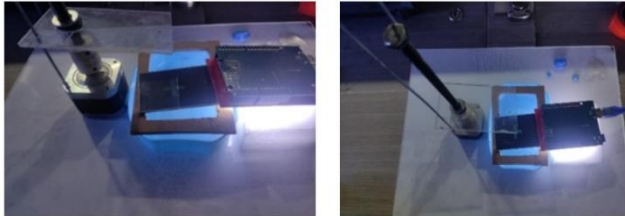


Figure 11. LCD Testing

2.6 Theoretical observations (Based on assumed values)

To calculate the depth of penetration (Dp) given the provided data and equations:

Given:

Wavelength: 365 nm

Critical Energy (Ec): 8.7 mJ/cm²

Cure depth (Cd): 1 cm.

Step 1: Convert the critical energy to J/mm²:

Critical Energy (Ec) = 8.7 mJ/cm² = 8.7 J/mm².....(10)

Step 2: Calculate the exposure time (t) using the formula.

$\text{Exp} = t \cdot \text{Irr}$: $\text{Exp} = t \cdot 8 \text{ W/m}^2 \cdot 3600 \text{ s/h}$

Since the lamp power (Irradiance) is constant at 8 W, we have:

$\text{Exp} = t \cdot 8 \text{ W/m}^2 \cdot 3600 \text{ s/h} = 28800 t \text{ Ws/m}^2$(11)

Step 3: Calculate the depth of penetration (Dp) using the formula

$\text{Cd} = \text{Dp} \cdot \ln (\text{Exp}) + \text{Ec}$:

Given that Cd = 1 cm = 10 mm, we can rearrange the formula as follows:

$\text{Dp} \cdot \ln (\text{Exp}) = \text{Cd} - \text{Ec}$

Substituting the values:

$\text{Dp} \cdot \ln (\text{Exp}) = 10 \text{ mm} - 8.7 \text{ J/mm}^2$(12)

Now, we can solve these equations to find the values of exposure time (t) and depth of penetration (Dp):

Using the equation $\text{Exp} = t \cdot \text{Irr}$, and since the irradiance (Irr) is constant, we have:

$\text{Exp} = 8 \text{ W/m}^2$

Plugging this into the equation $\text{Dp} \cdot \ln (\text{Exp}) = 10 \text{ mm} - 8.7 \text{ J/mm}^2$:

$\text{Dp} \cdot \ln (8) = 10 \text{ mm} - 8.7 \text{ J/mm}^2$(13)

Now, we can solve for Dp:

$\text{Dp} = (10 \text{ mm} - 8.7 \text{ J/mm}^2) / \ln (8)$

$\text{Dp} = (10 \text{ mm} - 8.7 \text{ J/mm}^2) / 2.079$

$\text{Dp} \approx 0.961 \text{ mm}$(14)

3 Results and Discussions

The Beer-Lambert relation is a fundamental principle in optics that describes the attenuation of light as it passes through a material. In the context of curing processes, it helps us understand how the energy from the UV lamp interacts with the curing material. The relation states that the amount of light absorbed by a material is directly proportional to the concentration of the absorbing species and the path length traversed by the light.

In this analysis, we considered a UV lamp with a wavelength of 365 nm and a constant irradiance of 8 W/m². The critical energy (Ec) for curing was determined to be 8.7 J/mm², and the cure depth (Cd) was given as 1 cm. Using the Beer-Lambert relation and relevant formulas, we calculated the depth of penetration (Dp) to be approximately 0.961 mm.

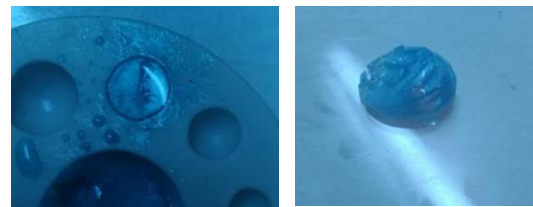


Figure 11. Resin Objects cured in our system.

The simulation of a resin 3D printer using Proteus was conducted to virtually model the Arduino board and its associated components like stepper motors and UV light sources. This process involved defining inputs and outputs, configuring the control interface, and establishing necessary circuit connections. By inputting simulated commands through the interface, the response of the Arduino board was observed, including its control over stepper motor movement, UV light source activation, and resin curing. The simulation allowed for assessment of movement accuracy, synchronization between components, and overall printing process coordination, ensuring the printer's functionality before physical implementation. [5][15][19][16]

Prototype 2, which cures resin within a wooden casing using a UV light tube, highlights the importance of

proper light focus for efficient resin hardening, along with safety considerations due to UV light exposure. Moreover, the printer's capabilities extend beyond prototyping to applications like resin embossing on items such as bottles and plates, showcasing its versatility for various industrial and creative purposes.

Prototype 3 reflects the shape displayed on the LCD module on the resin stored in the tank and thus cures the resin over time, although we had a few problems interfacing the stepper motor with the Arduino. The ball screw mechanism was very expensive and had to be outsourced hence a DIY ball screw was made for the mechanism of stepper motor to move the platform.

4 Conclusion

Prototype 1 is the simulation of the resin printer using Proteus, by inputting simulated commands through the control interface, it was observed how the Arduino board responds and controls the stepper motor's movement, UV light source activation, and resin curing process. One could note the accuracy of the movement, the synchronization between the platform and UV light source, and the overall coordination of the printing process.

Prototype 2 cures the resin in the wooden casing with the help of the UV light tube mounted inside the casing. The focus of the light must fall exactly on the curing resin for it to harden sooner.

Prototype 3 reflects the shape displayed on the LCD module on the resin stored in the tank and thus cures the resin over time.

Following this study, additive manufacturing has opened up a ton of new options for finishing manufacturing tasks. Stereolithography is one of the many diverse 3D printing methods that are available. Technology has advanced significantly. Thus, a low-cost 3D printer with superior surface finish, quick processing and manufacturing times, and short manufacturing cycles can be created. Resin will benefit from using the information because it employs an environmentally friendly process. Resin is a material that is sensitive to light and less toxic than thermoplastics such as ABS and other filament materials. [12][13][14]

In conclusion, our resin 3D printer's optimal depth of penetration (Dp) value of approximately 0.961 mm ensures efficient curing, faster printing speeds, and superior print quality. This advantage aligns perfectly with diverse application requirements, delivering high-quality prints with precision and consistency. Overall, these parameters are favorable, equipping our cost-effective printer to excel in various 3D printing applications.

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